

Internet Application Development

DST305

Prepared

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Methodology of working within this semester

- 1- the door will be closed after 15 minutes**
- 2- the last 15 minute in our lecture will be questions about the subject of our lecture.**
- 3- the final exam will include questions are similar to the questions of our lectures**
- 4- Each lecture has a quiz at the last 15 minute of our lecture, the quiz will be sent before the lecture.**
- 5- the absence is included in the quiz, each student has internet at the last 15 minute of our lecture.**
- 6- our marks will be classified as following:**
60 (med term) –15 (our lecture) – 15 (the lab) – 60 (final exam)
- 7- the marks for the lecture and lab are classified between absence, activities in lecture and assignment**
- 8- the total mark is 150 and the time for exam is 2 hours**

Some Major Definitions



1- Introduction

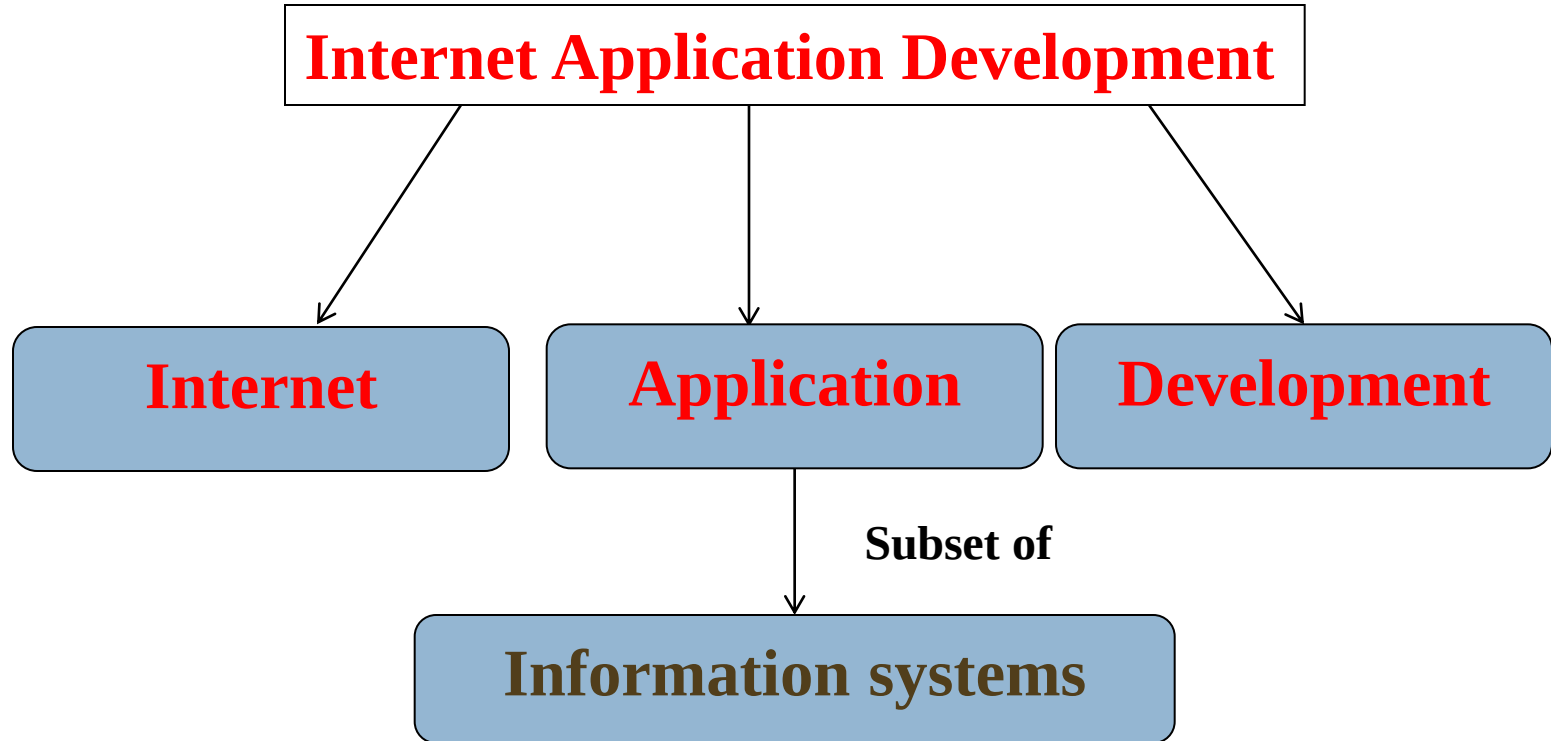
2- What is a definition of information system?

- ▣ What is a definition of system software ?
- ▣ What is a definition of application?
- ▣ what is an internet?
- ▣ What is a definition of web application?
- ▣ what is a difference between desktop and web application?

Internet Application Development

1- Introduction

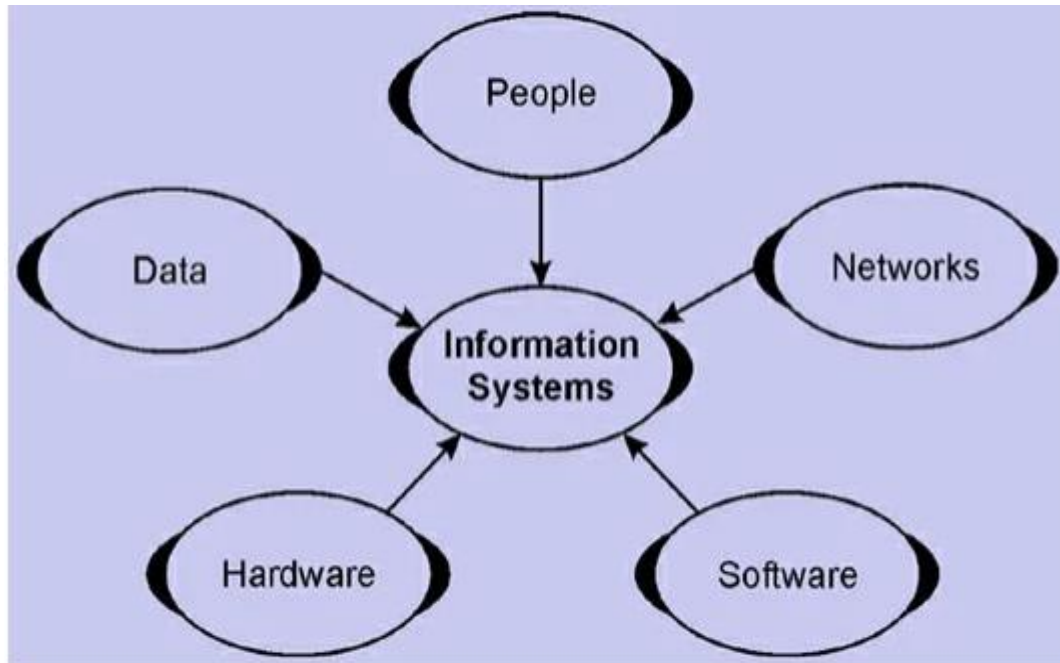
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2- What is a definition of information system?

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information system is a combination of software, hardware, and telecommunication networks and people to collect useful data, especially in an organization. Many businesses use information technology to complete and manage their operations, interact with their consumers, and stay ahead of their competition. Some companies today are completely built on information technology, such as eBay, Amazon, Alibaba, and Google.



2- What is a definition of information system?

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The components of information systems

It has **five** components – hardware, software, data, and telecommunications, people.

1. **Hardware** – it is the physical component of the technology. It includes computers, hard disks, keyboards, iPads, etc. The hardware **cost** has **decreased** rapidly while its speed and storage capacity has increased significantly.
2. **Software** – it is two types, system software and application software.
 - **The system software** is an operating system that manages the hardware, program files, and other resources while offering the user to control the PC using GUI.
 - **Application software** is designed to manage particular tasks by the users. In short, system software **makes** **the hardware usable** while **application software** handles specific tasks.



2- What is a definition of information system?

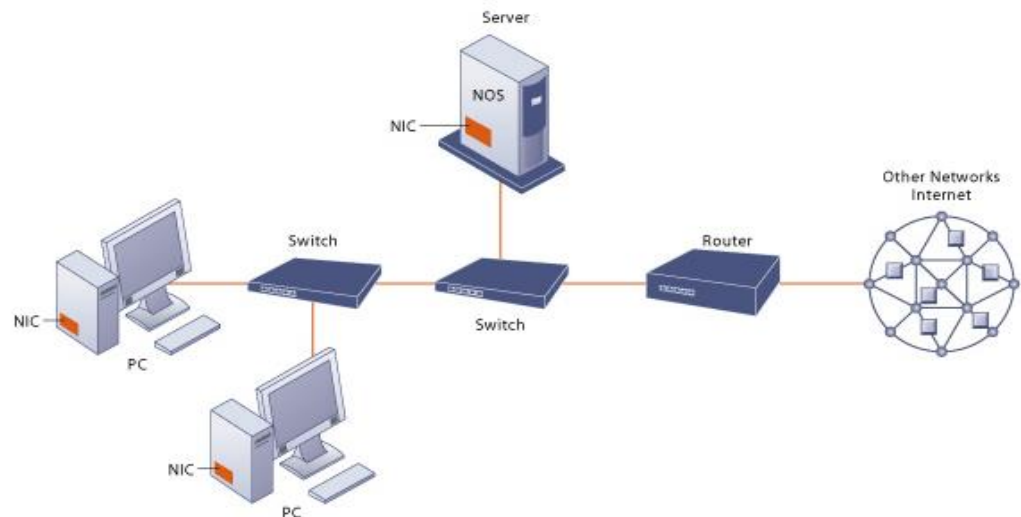
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The components of information systems

It has **five** components – hardware, software, data, and telecommunications, people.

3. Data : is a **collection** of **facts** and is useless by themselves, but when **collected** and organized together, it can be very **powerful** for business operations. Businesses collect all the data and **use** it to **make decisions** that can be **analyzed** for the **effectiveness** of the business operations.

4. Telecommunications: is used to connect with the computer system or other devices to **disseminate** information. The **network** can be **established** using **wired** or **wireless** modes. Wired technologies include fiber optics and coaxial cable, while wireless technologies include radio waves and microwaves.



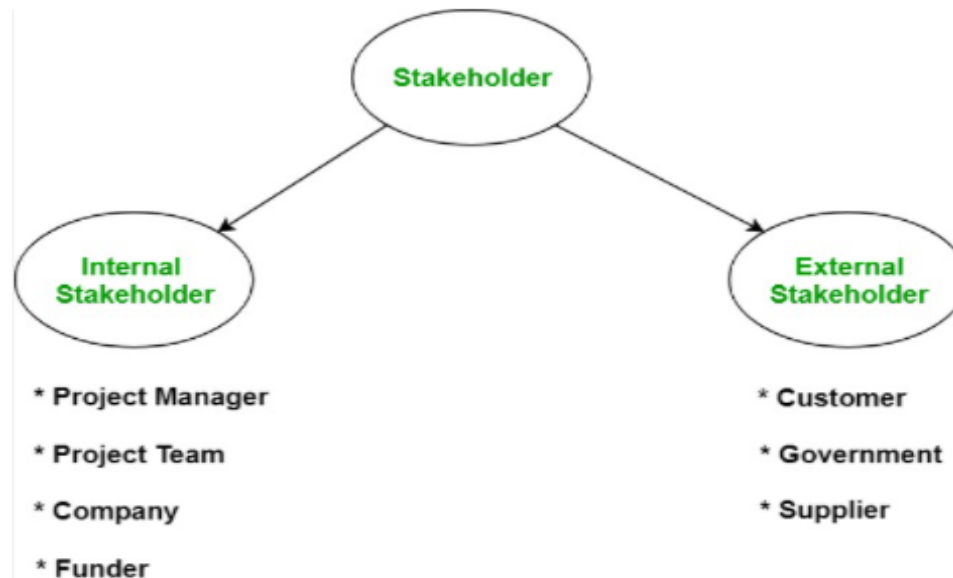
2- What is a definition of information system?

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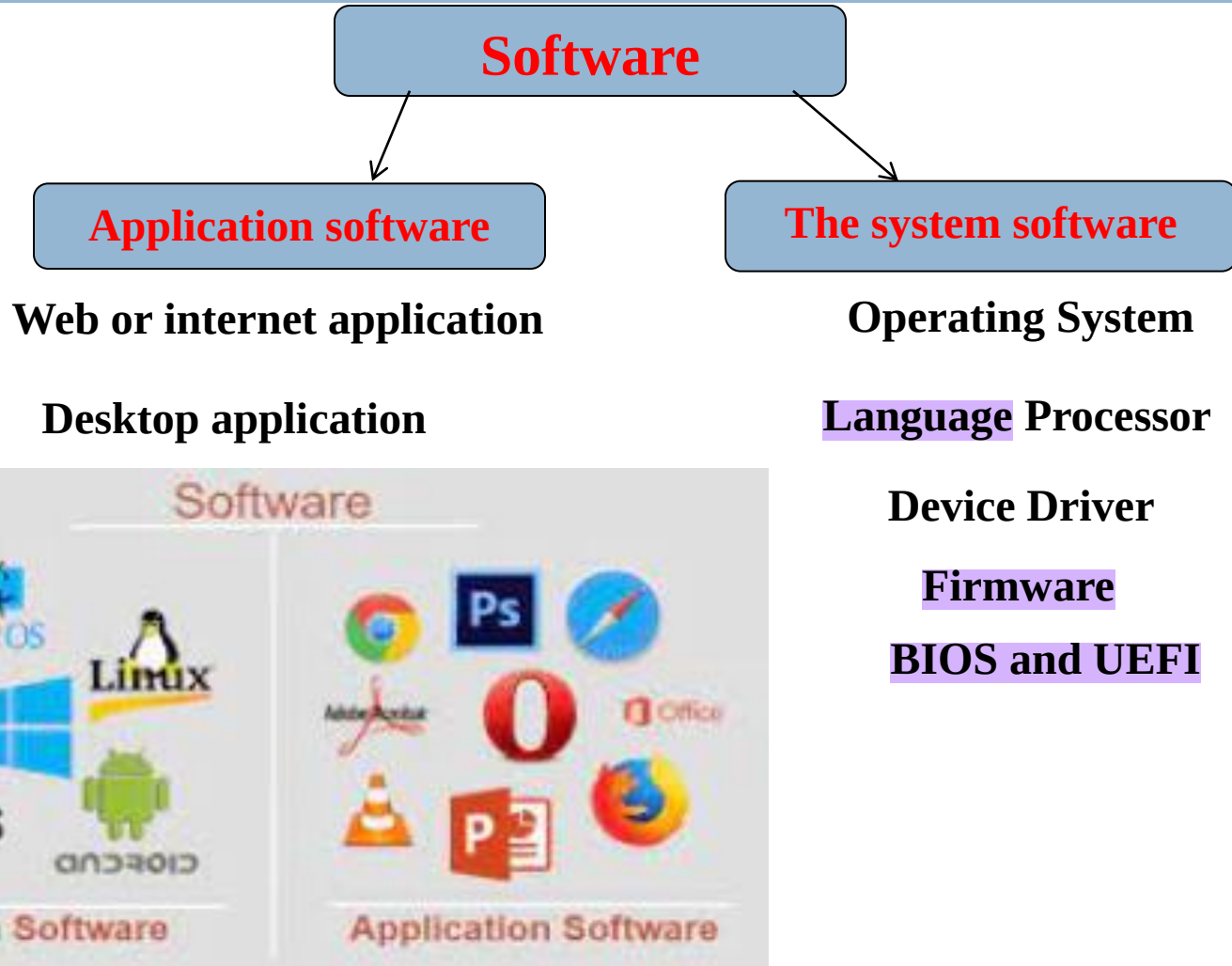
The components of information systems

It has five components – hardware, software, data, and telecommunications, people.

5. People : is any one can affect and can be affected in information system. This is called a **stakeholder**. This includes individuals, groups, organizations, and nations. The primary **stakeholders** in information systems **are** the **users** of the system. They are the ones who will be most affected by the system and who will have the most to **gain** or **lose** from its success or failure. **Other** stakeholders include the system **developers**, the system **maintainers**, and the **system administrators**.



2- What is a definition of information system?



2- What is a definition of information system?

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System software is the interface between application software and the system. It maintains the system resources and gives the path for application software to run. Low-level languages are used to write the system software. The application software can not run without system software.



2- What is a definition of information system?

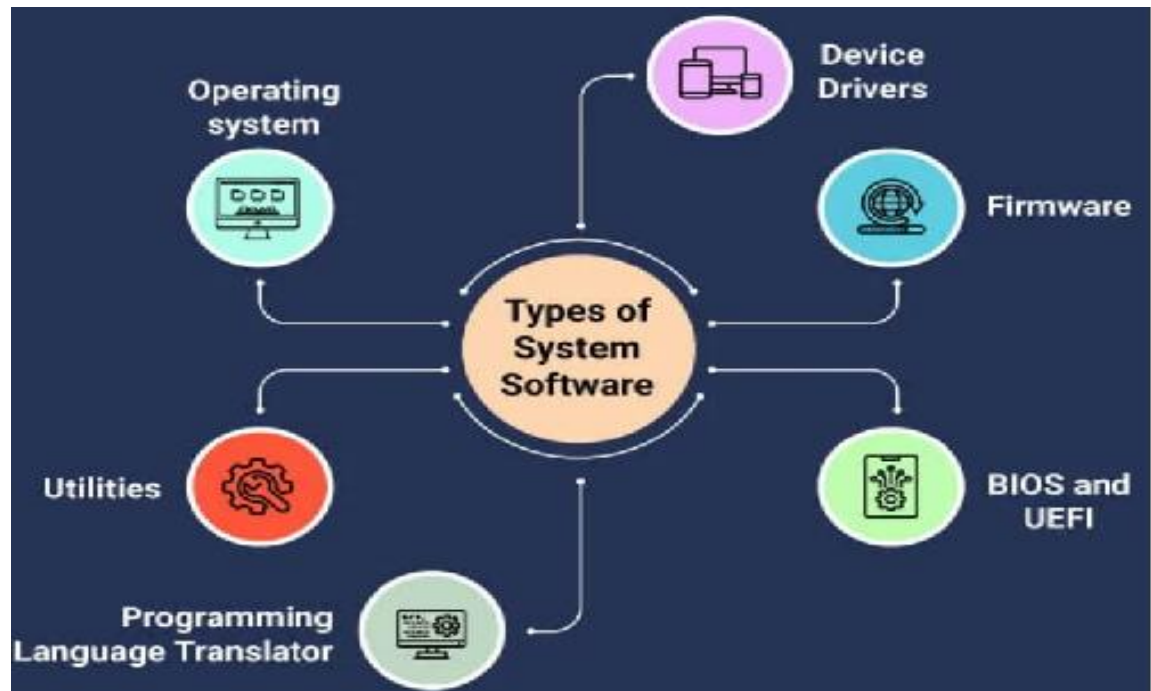
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Functions of System Software

1. Memory Management
2. Processor Management
3. File Management
4. Security
5. Error-detecting Aids
6. Scheduling

Features of System Software

1. It is written in a low-level language.
2. The size of the system Software is smaller.
3. It is complex to understand.
4. It presents near hardware components.



2- What is a definition of information system?

Types of System Software

Operating System: is the main part of the Computer System. It manages all the resources such as CPU, Printer, Hard Disk, etc. Examples of Operating Systems are Linux, Apple, macOS, Microsoft Windows, etc.

Language Processor: it converts Human-Readable Language into a Machine Language. It converts programs into instructions that are easily readable by Machines.

Device Driver: is a program or software performs its functions by controlling the device. It is installed firstly as a driver for running the program.

Firmware: is an operational software stored in a flash, ROM, EPROM for the OS to identify it. It provides instructions on how the device should be operated. Unlike other software, firmware can not be manipulated, changed, or deleted by the end-users. They remain on the device.

BIOS and UEFI:

BIOS (Basic Input/Output System) or the new UEFI (Unified Extended Firmware Software) gets the computer system started when we turn it on. BIOS also manages the flow of information between operating systems and the attached devices.

2- What is a definition of information system?

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It is a computer software package which performs a specific function directly for an end user or for another application. **An application** can be self-contained or a group of programs. The program is a set of operations that runs the application for the user. In addition, it has two names (an application program or application software)

Application is such as Word processors, Database programs, Web browsers, Deployment tools, Image editors, Communication platforms, spreadsheets, database management, inventory and payroll programs.



2- What is a definition of information system?

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An application, also referred to as an application program or application software, is a computer software package that performs a specific function directly for an end user or, in some cases, for another application. An application can be self-contained or a group of programs.



2- What is a definition of information system?

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Desktop application:

is a program or software that run locally on a computer after installation. It takes up space on the hard drive of the computer and generally run without an internet connection. It is considered a traditional application because the companies started their information system with this type of application. For example, Microsoft Office applications, Adobe Illustrator, Adobe Photoshop, media players, etc.



2- What is a definition of information system?

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Web or internet application:

Is not required to be installed in the computer and is accessible within web browsers in the presence of an internet connection. Contrarily to desktop apps, web applications take up space on the server. As most web applications are designed to be responsive, and interactive, and can run on computers, mobile devices, and all other devices with a web browser, it is considered to be the modern form of computer applications and their use is massively growing day by day. For example, Google docs, YouTube, Facebook, Netflix, etc. are a few examples of web applications.



2- What is a definition of information system?

The Internet is a worldwide interconnected network of hundreds of thousands of computers of various types that belong to multiple networks.

It uses an Internet Protocol (IP) and Transport Control Protocol (TCP)-based packet routing network to ensure that data transmission across the internet is consistent and reliable, regardless of the device or location.

Uses of the Internet:

E-mail: is an electronic message is sent across a network from one computer user to one or more recipients and an electronic message is received by servers.

Web Chat: Web chat is an application that allows you to send and receive messages in real-time with others.

Video conferencing: it refers to the use of voice and images to communicate amongst users.

World Wide Web: The World Wide Web is the Internet's most popular information exchange service using hypertext or hyperlinks.



2- What is a definition of information system?

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Uses of the Internet:

E-commerce: it is electronic business transactions made over the Internet such as Amazon, Flipkart, and Myntra.

Cashless Transactions: All the merchandising companies are offering services to their customers to pay the bills of the products online via various digital payment apps such as Paytm, Google Pay, etc.

Education: It refers to E-learning. It enables students to study online who are unable to attend physical classes can choose any course from the internet and can have point-to-point knowledge of it just by sitting at home.

Social Networking: it uses to connect people all over the world and can talk, share videos, and images with other ones. Also, people can create groups for discussion or for meetings.

Internet telephony: The technique that converts analog speech impulses into digital signals and routes them through packet-switched networks of the internet.

Entertainment: There are many entertainment options available on the internet such as watching movies, playing games, listening to music, etc. people can download movies, games, songs, TV Serial, etc., easily from the internet.

2- What is a definition of information system?

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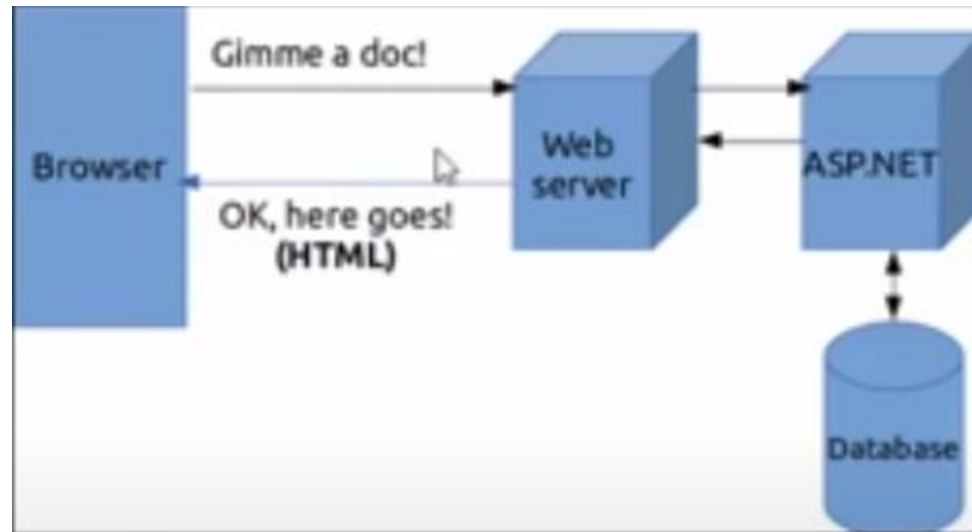
DESKTOP APPS	WEB APPS
They require installation on the computer to run.	They are accessible through web browsers and do not require installation.
Generally, desktop apps do not require an internet connection to run.	Web apps cannot run without an internet connection.
They are accessible only in the machine they are installed in.	They are accessible from anywhere and through any device with an internet connection and a web browser.
They take space on the hard drive of the local computer.	They take up space on the remote server.
Deployment and updating are to be done individually on each computer.	Deployment and updating are done only on the server.
They have strict hardware requirements for proper functionality.	Web apps are hardware-independent and just require a web browser and internet connection to function.
As they are confined to a device and single or limited users, they are highly secure.	As web apps are accessible to all through the internet, they are less secure than desktop apps.
Generally, they are faster than web applications.	Generally, they are slower than desktop applications.

2- What is a definition of information system?

The difference between website and web application?

Website is a set of static web pages to show specific thing without interactive with the user within the browser. It contains of multimedia

Web application is a set of interactive web pages with the user. The user requests



2- What is a definition of information system?

The prime difference between web application and web site

Parameter	Web application	Website
Created for	A web application is designed for interaction with the end user	A website mostly consists of static content. It is publicly accessible to all the visitors.
User interaction	In a web application, the user not only read the page content but also manipulate the restricted data.	A website provides visual & text content which user can view and read, but not affect it's functioning.
Authentication	Web applications need authentication, as they offer a much broader scope of options than websites.	Authentication is not obligatory for informational websites. The user may ask to register to get a regular update or to access additional options. This features not available for the unregistered website visitors.

2- What is a definition of information system?

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The prime difference between web application and web site

Parameter	Web application	Website
Task and Complexity	Web application functions are quite higher and complex compared to a website.	The website displays the collected data and information on a specific page.
Type of software	The web application development is part of the website. It is itself not a complete website.	The website is a complete product, which you access with the help of your browser.
Compilation	The site must be precompiled before deployment	The site doesn't need to be pre-compiled
Deployment	All changes require the entire project to be re-compiled and deployed.	Small changes never require a full re-compilation and deployment. You just need to update the HTML code.

2- What is a definition of information system?

The Disadvantages of web application and web site

Web application	Website
Security is not guaranteed, so it is vulnerable for unauthorized access.	A website can crash which is not good for anyone. It is the biggest disadvantage for your business
The web app may not support multiple browsers with equal precedence.	Contact form published on your website may invite lots of unwanted spam e-mails.
The web application is built explicitly for a certain operating system, so it is difficult to discover from the app store.	The information on any website might be unreliable if it is not updated regularly.
Limited scope to access the device's features.	

An examples of web application and web site

Banks, airline, face book	All static websites such as personal webs ...etc
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Internet Application Development Software Development Life Cycle (SDLC)

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It is a model represents all the stages which are required to make a software product. It also captures the structure in which these methods are to be undertaken.